

# TradingAgents Benchmark Report

**Run ID:** bt\_20260129\_173611 **Date:** 2026-02-02 **Period:** 2025-01-28 to 2026-01-28 (12 months)  
**Tickers:** RAPT, SUPN, ZTS, SNOW, AMBA, ET (6 tickers) **Monthly Snapshots:** 12 (last trading day of each month) **Strategies Tested:** 7 (4 multi-agent, 3 single-agent) **Total Analyses:** 504 completed, 0 failed

## Configuration

| Setting                    | Value                       |
|----------------------------|-----------------------------|
| LLM Provider               | OpenAI                      |
| Memory (ChromaDB)          | Enabled                     |
| ACE Framework              | Disabled                    |
| Max Debate Rounds          | 1                           |
| Max Risk Discussion Rounds | 1                           |
| News Lookback              | 90 days per analysis window |
| Articles per Month         | 50                          |

## Data Sources

| Category             | Primary Vendor                                         | Fallback                 |
|----------------------|--------------------------------------------------------|--------------------------|
| Stock Prices (OHLCV) | Polygon.io                                             | Alpha Vantage, yfinance  |
| Technical Indicators | Polygon.io (SMA/EMA/RSI/MACD)                          | yfinance for unsupported |
| Fundamentals         | Alpha Vantage                                          | yfinance                 |
| Balance Sheet        | Alpha Vantage                                          | yfinance                 |
| Cash Flow            | Alpha Vantage                                          | yfinance                 |
| Income Statement     | Alpha Vantage                                          | yfinance                 |
| News                 | Polygon.io + Alpha Vantage (dual-source, deduplicated) | Google, local            |
| Social Sentiment     | Stocktwits (primary)                                   | Finnhub, Reddit PRAW     |

- **Polygon.io cache:** 571 cached data files (stock prices, indicators, news)
- **Social sentiment cache:** 30 cached files (Stocktwits messages)
- **Analyst reports generated:** 77 per report type (market, sentiment, news, fundamentals) across 6 tickers x ~13 months

- **Report caching:** Multi-agent analyst reports (market, sentiment, news, fundamentals) are generated once per ticker/date, then shared across all multi-agent strategy variants

Model Configurations Tested

| Strategy              | Type                      | Deep Think Model   | Quick Think Model  |
|-----------------------|---------------------------|--------------------|--------------------|
| ma_5.2+5.1-codex-mini | Multi-agent (12 agents)   | gpt-5.2-2025-12-11 | gpt-5.1-codex-mini |
| ma_5.1-codex-mini     | Multi-agent (12 agents)   | gpt-5.1-codex-mini | gpt-5.1-codex-mini |
| ma_5.2                | Multi-agent (12 agents)   | gpt-5.2-2025-12-11 | gpt-5.2-2025-12-11 |
| ma_4o                 | Multi-agent (12 agents)   | gpt-4o             | gpt-4o             |
| single_5.1-codex-mini | Single-agent (1 LLM call) | gpt-5.1-codex-mini | —                  |
| single_5.2            | Single-agent (1 LLM call) | gpt-5.2-2025-12-11 | —                  |
| single_4o             | Single-agent (1 LLM call) | gpt-4o             | —                  |

**Multi-agent pipeline:** 4 Analysts (Market, Social, News, Fundamentals) → 2 Researchers (Bull, Bear) → 1 Trader → 2 Risk Managers, with bull/bear debate and risk discussion rounds.

**Single-agent baseline:** One LLM call with all raw data (market, indicators, fundamentals, balance sheet, cash flow, income, news, global news, sentiment) passed directly in context.

Executive Summary

| Metric                | single_5.1-codex-mini | single_5.2 | ma_5.2+5.1-codex-mini | ma_5.2 | ma_5.1-codex-mini | ma_4o     | single_4o  |
|-----------------------|-----------------------|------------|-----------------------|--------|-------------------|-----------|------------|
| Overall accuracy      | 45.8%                 | 40.3%      | 38.9%                 | 40.3%  | 33.3%             | 43.1%     | 40.3%      |
| Total P&L (post-cost) | +\$1,173.32           | +\$519.09  | +\$26.32              | \$0.00 | -\$389.58         | -\$408.82 | -\$759.35  |
| Max drawdown          | \$629.64              | \$624.83   | \$142.47              | \$0.00 | \$389.58          | \$939.10  | \$1,367.10 |
| Avg confidence        | 0.37                  | 0.64       | 0.60                  | 0.59   | 0.52              | 0.52      | 0.53       |
| Active trades         | 21 (29%)              | 9 (12%)    | 3 (4%)                | 0 (0%) | 9 (12%)           | 4 (6%)    | 7 (10%)    |
| Win rate (on trades)  | 57%                   | 44%        | 33%                   | N/A    | 11%               | 50%       | 29%        |
| Avg win               | +\$220.84             | +\$359.40  | +\$168.79             | —      | +\$272.04         | +\$265.14 | +\$458.83  |

| Metric   | single_5.1-codex-mini | single_5.2 | ma_5.2+5.1-codex-mini | ma_5.2 | ma_5.1-codex-mini | ma_4o     | single_4o |
|----------|-----------------------|------------|-----------------------|--------|-------------------|-----------|-----------|
| Avg loss | -\$164.09             | -\$183.70  | -\$71.23              | —      | -\$82.70          | -\$469.55 | -\$335.40 |
| BUY %    | 3%                    | 0%         | 0%                    | 0%     | 3%                | 0%        | 1%        |
| SELL %   | 26%                   | 12%        | 4%                    | 0%     | 10%               | 6%        | 8%        |
| HOLD %   | 71%                   | 88%        | 96%                   | 100%   | 88%               | 94%       | 90%       |
| Analyses | 72                    | 72         | 72                    | 72     | 72                | 72        | 72        |

Single-Agent vs Multi-Agent (Aggregated)

| Category                               | Accuracy      | P&L       | Active Trade Rate |
|----------------------------------------|---------------|-----------|-------------------|
| Single-agent (3 strategies, 216 tasks) | 42% (91/216)  | +\$933.06 | 17% (37 trades)   |
| Multi-agent (4 strategies, 288 tasks)  | 39% (112/288) | -\$772.08 | 6% (16 trades)    |

Ticker Difficulty Analysis

Tickers ranked from hardest to easiest based on average accuracy across all 7 strategies:

| Rank           | Ticker | Avg Accuracy | Aggregate P&L | Best Strategy              | Worst Strategy                         | Notes                                                                                                                                                                               |
|----------------|--------|--------------|---------------|----------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1<br>(hardest) | RAPT   | 10%          | +\$334        | single_5.2<br>(+\$808, 8%) | ma_4o<br>(-\$614, 8%)                  | Micro-cap biotech (\$9-\$12 range). Extreme volatility, very low news coverage. Most strategies scored only 1/12 correct. Only profits came from rare correct SELL calls.           |
| 2              | SNOW   | 27%          | +\$49         | single_4o<br>(+\$681, 42%) | single_5.1-codex-mini<br>(-\$367, 33%) | High-growth tech. Inconsistent performance — the best overall strategy (single_5.1-codex-mini) actually lost money here. Market dynamics too unpredictable for any consistent edge. |

| Rank           | Ticker | Avg Accuracy | Aggregate P&L | Best Strategy                       | Worst Strategy                   | Notes                                                                                                                                                                                          |
|----------------|--------|--------------|---------------|-------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3              | AMBA   | 32%          | -\$401        | single_5.1-codex-mini (+\$445, 33%) | single_4o (-\$739, 25%)          | Semiconductor. Wide dispersion between strategies — winner made +\$445, loser lost -\$739. GPT-4o particularly poor on this ticker.                                                            |
| 4              | SUPN   | 54%          | +\$763        | single_5.1-codex-mini (+\$679, 75%) | ma_5.2+5.1-codex-mini (\$0, 50%) | Specialty pharma with clearer directional signals. Best ticker for the winning strategy (75% accuracy). Multi-agent strategies defaulted to HOLD, missing profitable opportunities.            |
| 5              | ZTS    | 56%          | -\$562        | ma_5.2+5.1-codex-mini (\$0, 58%)    | ma_5.1-codex-mini (-\$256, 50%)  | Large-cap animal health. High accuracy but negative P&L — strategies predicted direction correctly but lost on the trades they did make. Safe for HOLD strategies.                             |
| 6<br>(easiest) | ET     | 63%          | -\$22         | single_5.1-codex-mini (+\$105, 67%) | single_4o (-\$90, 58%)           | Energy midstream/MLP. Most predictable — stable price action, high dividend yield making HOLD often correct. Near break-even aggregate P&L despite high accuracy because few trades were made. |

## Key Observations on Ticker Behavior

- **Accuracy does not equal profitability:** ZTS had 56% average accuracy but -\$562 aggregate P&L. ET had 63% accuracy but -\$22 P&L. High accuracy on HOLD-dominated strategies doesn't generate returns.
- **RAPT's 10% accuracy** across 84 tasks means the market moved against virtually every prediction. This is a biotech micro-cap with binary event risk (clinical trials, FDA decisions) that no amount of financial data can predict.
- **SUPN was the profit engine** — +\$763 aggregate P&L, driven primarily by single\_5.1-codex-mini's 75% accuracy and profitable SELL calls.

## Active Trading Analysis

### Win Rate When Actually Trading (non-HOLD Only)

| Strategy                     | Total Trades | Wins      | Losses   | Win Rate   | Avg Win P&L | Avg Loss P&L | Win/Loss Ratio |
|------------------------------|--------------|-----------|----------|------------|-------------|--------------|----------------|
| <b>single_5.1-codex-mini</b> | <b>21</b>    | <b>12</b> | <b>9</b> | <b>57%</b> | +\$220.84   | -\$164.09    | <b>1.35:1</b>  |
| single_5.2                   | 9            | 4         | 5        | 44%        | +\$359.40   | -\$183.70    | 1.96:1         |
| ma_4o                        | 4            | 2         | 2        | 50%        | +\$265.14   | -\$469.55    | 0.56:1         |
| ma_5.2+5.1-codex-mini        | 3            | 1         | 2        | 33%        | +\$168.79   | -\$71.23     | 2.37:1         |
| ma_5.1-codex-mini            | 9            | 1         | 8        | <b>11%</b> | +\$272.04   | -\$82.70     | 3.29:1         |
| single_4o                    | 7            | 2         | 5        | 29%        | +\$458.83   | -\$335.40    | 1.37:1         |
| ma_5.2                       | <b>0</b>     | —         | —        | N/A        | —           | —            | —              |

### HOLD-Only Accuracy (Was Doing Nothing Correct?)

| Strategy              | Correct HOLDS | HOLD Accuracy |
|-----------------------|---------------|---------------|
| ma_4o                 | 29/68         | 43%           |
| single_4o             | 27/65         | 42%           |
| single_5.1-codex-mini | 21/51         | 41%           |
| ma_5.2                | 29/72         | 40%           |
| single_5.2            | 25/63         | 40%           |
| ma_5.2+5.1-codex-mini | 27/69         | 39%           |
| ma_5.1-codex-mini     | 23/63         | 37%           |

HOLD accuracy is ~40% across all strategies, meaning the market moved meaningfully ~60% of the time. Strategies with very high HOLD rates (ma\_5.2 at 100%, ma\_5.2+5.1-codex-mini at 96%) are systematically leaving money on the table.

Results by Month

| Month   | single_5.1-codex-mini | single_5.2 | ma_5.2+5.1-codex-mini | ma_5.2    | ma_5.1-codex-mini | ma_4o     | single_4o |
|---------|-----------------------|------------|-----------------------|-----------|-------------------|-----------|-----------|
| 2025-01 | 2/6 (33%)             | 2/6 (33%)  | 4/6 (67%)             | 3/6 (50%) | 2/6 (33%)         | 3/6 (50%) | 3/6 (50%) |
| 2025-02 | 3/6 (50%)             | 1/6 (17%)  | 1/6 (17%)             | 1/6 (17%) | 1/6 (17%)         | 1/6 (17%) | 1/6 (17%) |
| 2025-03 | 2/6 (33%)             | 1/6 (17%)  | 2/6 (33%)             | 2/6 (33%) | 2/6 (33%)         | 2/6 (33%) | 1/6 (17%) |
| 2025-04 | 4/6 (67%)             | 4/6 (67%)  | 4/6 (67%)             | 4/6 (67%) | 3/6 (50%)         | 4/6 (67%) | 4/6 (67%) |
| 2025-05 | 4/6 (67%)             | 3/6 (50%)  | 4/6 (67%)             | 4/6 (67%) | 3/6 (50%)         | 4/6 (67%) | 3/6 (50%) |
| 2025-06 | 1/6 (17%)             | 2/6 (33%)  | 1/6 (17%)             | 1/6 (17%) | 1/6 (17%)         | 1/6 (17%) | 1/6 (17%) |
| 2025-07 | 1/6 (17%)             | 0/6 (0%)   | 0/6 (0%)              | 0/6 (0%)  | 1/6 (17%)         | 0/6 (0%)  | 1/6 (17%) |
| 2025-08 | 1/6 (17%)             | 1/6 (17%)  | 1/6 (17%)             | 1/6 (17%) | 1/6 (17%)         | 1/6 (17%) | 1/6 (17%) |
| 2025-09 | 2/6 (33%)             | 3/6 (50%)  | 3/6 (50%)             | 3/6 (50%) | 3/6 (50%)         | 3/6 (50%) | 3/6 (50%) |
| 2025-10 | 4/6 (67%)             | 4/6 (67%)  | 3/6 (50%)             | 4/6 (67%) | 1/6 (17%)         | 4/6 (67%) | 4/6 (67%) |
| 2025-11 | 5/6 (83%)             | 4/6 (67%)  | 2/6 (33%)             | 3/6 (50%) | 3/6 (50%)         | 4/6 (67%) | 3/6 (50%) |
| 2025-12 | 4/6 (67%)             | 4/6 (67%)  | 3/6 (50%)             | 3/6 (50%) | 3/6 (50%)         | 4/6 (67%) | 4/6 (67%) |

Cumulative P&L: single\_5.1-codex-mini (Best Strategy)

| Month   | Monthly P&L | Cumulative P&L | Accuracy  |
|---------|-------------|----------------|-----------|
| 2025-01 | -\$15.00    | -\$15.00       | 2/6 (33%) |
| 2025-02 | +\$941.67   | +\$926.67      | 3/6 (50%) |
| 2025-03 | +\$329.16   | +\$1,255.84    | 2/6 (33%) |
| 2025-04 | +\$55.07    | +\$1,310.91    | 4/6 (67%) |

| Month   | Monthly P&L | Cumulative P&L | Accuracy  |
|---------|-------------|----------------|-----------|
| 2025-05 | \$0.00      | +\$1,310.91    | 4/6 (67%) |
| 2025-06 | \$0.00      | +\$1,310.91    | 1/6 (17%) |
| 2025-07 | -\$22.25    | +\$1,288.66    | 1/6 (17%) |
| 2025-08 | -\$227.02   | +\$1,061.64    | 1/6 (17%) |
| 2025-09 | +\$26.01    | +\$1,087.65    | 2/6 (33%) |
| 2025-10 | \$0.00      | +\$1,087.65    | 4/6 (67%) |
| 2025-11 | +\$604.69   | +\$1,692.34    | 5/6 (83%) |
| 2025-12 | -\$519.02   | +\$1,173.32    | 4/6 (67%) |

Most profits came from Q1 2025 (+\$1,256 in first 3 months) with a second surge in Nov 2025 (+\$605). Mid-year (Jun-Aug) was a drawdown period across all strategies.

## Results by Ticker

| Ticker | single_5.1-codex-mini | single_5.2   | ma_5.2+5.1-codex-mini | ma_5.2    | ma_5.1-codex-mini | ma_4o        | single_4o           |
|--------|-----------------------|--------------|-----------------------|-----------|-------------------|--------------|---------------------|
| RAPT   | 17% / +\$494          | 8% / +\$808  | 8% / \$0              | 8% / \$0  | 8% / +\$257       | 8% / -\$614  | 8% / -\$611         |
| SUPN   | <b>75% / +\$679</b>   | 50% / +\$85  | 50% / \$0             | 50% / \$0 | 50% / \$0         | 50% / \$0    | 50% / \$0           |
| ZTS    | 50% / -\$182          | 58% / -\$124 | 58% / \$0             | 58% / \$0 | 50% / -\$256      | 58% / \$0    | 58% / \$0           |
| SNOW   | 33% / -\$367          | 33% / +\$106 | 8% / -\$142           | 25% / \$0 | 17% / -\$334      | 33% / +\$106 | <b>42% / +\$681</b> |
| AMBA   | 33% / +\$445          | 25% / -\$355 | <b>42% / +\$169</b>   | 33% / \$0 | 25% / -\$21       | 42% / +\$99  | 25% / -\$739        |
| ET     | <b>67% / +\$105</b>   | 67% / \$0    | 67% / \$0             | 67% / \$0 | 50% / -\$36       | 67% / \$0    | 58% / -\$90         |

## Decision Distribution

| Strategy              | BUY    | SELL            | HOLD     | Active Rate |
|-----------------------|--------|-----------------|----------|-------------|
| single_5.1-codex-mini | 3% (2) | <b>26% (19)</b> | 71% (51) | <b>29%</b>  |
| single_5.2            | 0% (0) | 12% (9)         | 88% (63) | 12%         |
| ma_5.1-codex-mini     | 3% (2) | 10% (7)         | 88% (63) | 12%         |

| Strategy              | BUY    | SELL   | HOLD      | Active Rate |
|-----------------------|--------|--------|-----------|-------------|
| single_4o             | 1% (1) | 8% (6) | 90% (65)  | 10%         |
| ma_4o                 | 0% (0) | 6% (4) | 94% (68)  | 6%          |
| ma_5.2+5.1-codex-mini | 0% (0) | 4% (3) | 96% (69)  | 4%          |
| ma_5.2                | 0% (0) | 0% (0) | 100% (72) | 0%          |

Confidence Analysis

| Strategy              | Avg Confidence | High-Conf (>=0.6) Accuracy | Low-Conf (<0.6) Accuracy | Parseable    |
|-----------------------|----------------|----------------------------|--------------------------|--------------|
| single_5.2            | 0.64           | 41% (70 trades)            | 0% (2 trades)            | 72/72 (100%) |
| ma_5.2+5.1-codex-mini | 0.60           | 32% (44 trades)            | 50% (28 trades)          | 72/72 (100%) |
| ma_5.2                | 0.59           | 50% (36 trades)            | 31% (36 trades)          | 72/72 (100%) |
| single_4o             | 0.53           | 33% (33 trades)            | 46% (39 trades)          | 72/72 (100%) |
| ma_4o                 | 0.52           | 33% (9 trades)             | 44% (63 trades)          | 72/72 (100%) |
| ma_5.1-codex-mini     | 0.52           | 0% (4 trades)              | 35% (68 trades)          | 72/72 (100%) |
| single_5.1-codex-mini | 0.37           | 100% (1 trade)             | 45% (69 trades)          | 70/72 (97%)  |

**Confidence calibration paradox:** `single_5.1-codex-mini` has the *lowest* average confidence (0.37) but the *highest* accuracy (46%). It is under-confident but correct more often. Conversely, `single_5.2` has the highest confidence (0.64) with lower accuracy (40%). This suggests `gpt-5.1-codex-mini` is better calibrated — it knows what it doesn't know — while `gpt-5.2` is overconfident.

Data Sources Detail

Stock Price Data (Polygon.io)

- **Data type:** Daily OHLCV (Open, High, Low, Close, Volume)
- **History window:** 90 days prior to each analysis date
- **Indicators computed:** SMA (20/50/200-day), EMA (12/26-day), RSI (14-day), MACD (12/26/9)
- **Coverage:** All 6 tickers fully covered for the full 12-month test period
- **Cache:** 571 Polygon data files cached locally to avoid redundant API calls

Fundamental Data (Alpha Vantage)



- **Data types:** Company overview, balance sheet, cash flow statement, income statement
- **Coverage:** Quarterly financials for all 6 tickers
- **Report size:** Avg 7.8 KB per fundamentals report (detailed financial metrics)

### News Data (Polygon.io + Alpha Vantage, dual-source)

- **Sources:** Polygon.io news API + Alpha Vantage news API with cross-source deduplication
- **Lookback:** 90 days of news per analysis window
- **Article limit:** 50 articles per analysis
- **Deduplication:** Jaccard similarity (0.7 threshold) on article titles to remove cross-source duplicates
- **Company name fallback:** Enabled — retries with company name if ticker search yields few results (useful for small-cap tickers like RAPT)
- **Report size:** Avg 6.8 KB per news report, avg 6.5 KB per market report

### Social Sentiment Data (Stocktwits)

- **Primary source:** Stocktwits (free, no API key required)
- **Data type:** User messages with bullish/bearish/neutral sentiment labels
- **Cache:** 30 sentiment cache files across all ticker/date combinations
- **Report size:** Avg 2.0 KB per sentiment report
- **Note:** ApeWisdom was disabled for this run (unreliable data quality). Finnhub and Reddit PRAW available as optional fallback sources.

### Analyst Reports (LLM-generated, cached per ticker/date)

- **Market report:** Technical analysis, price levels, trend assessment (avg 6.5 KB)
- **Sentiment report:** Social media sentiment summary (avg 2.0 KB)
- **News report:** News digest with macro/micro analysis (avg 6.8 KB)
- **Fundamentals report:** Valuation, earnings quality, balance sheet health (avg 7.8 KB)
- **Final trade decision:** Multi-agent consensus decision (avg 3.1 KB)
- **Total:** 77 reports per type, 385+ reports total across all categories
- **Caching:** Reports generated by the first multi-agent strategy per ticker/date, then reused by all subsequent multi-agent variants (only debate/trader/risk layers re-run with different models)

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## Conclusions

### 1. Single-Agent Decisively Outperforms Multi-Agent

Across 504 analyses, single-agent strategies generated **+\$933 aggregate P&L at 42% accuracy**, while multi-agent strategies lost **-\$772 at 39% accuracy**. This is not marginal — it's a reversal. The 12-agent pipeline (4 analysts, 2 researchers, 1 trader, 2 risk managers) actively destroys the alpha that individual models can generate.

### 2. The Multi-Agent Debate Process Kills Conviction

The core problem is systematic: the bull/bear debate followed by risk manager review collapses almost every actionable signal into HOLD. Evidence:

- **ma\_5.2** made **zero trades** across 72 tasks (100% HOLD)

- **ma\_5.2+5.1-codex-mini** made only 3 trades (96% HOLD)
- Multi-agent active trade rate: **6%** vs single-agent **17%**
- When multi-agent strategies *did* trade, they performed poorly (ma\_5.1-codex-mini: 11% win rate)

The consensus mechanism is a bug, not a feature. It doesn't filter bad trades — it filters *all* trades.

### 3. GPT-5.1-codex-mini Is the Best Model for Trading Decisions

**gpt-5.1-codex-mini** as a single agent achieved:

- **Highest accuracy:** 45.8% (33/72 correct)
- **Highest P&L:** +\$1,173.32
- **Most active trading:** 21 trades (29% active rate)
- **Best win rate:** 57% on active trades
- **Favorable risk/reward:** 1.35:1 win/loss ratio (\$221 avg win vs \$164 avg loss)
- **Best calibration:** Lowest confidence (0.37) but highest accuracy — knows what it doesn't know

### 4. GPT-4o Is the Weakest Model

GPT-4o produced negative P&L in both configurations:

- **single\_4o:** -\$759, 40% accuracy, 29% win rate on trades
- **ma\_4o:** -\$409, 43% accuracy, 50% win rate but massive avg losses (\$470)

GPT-4o's problem is not accuracy (40-43% is competitive) but *trade quality* — when it acts, its losses are disproportionately large compared to wins.

### 5. Model Pairing Doesn't Help Multi-Agent

The mixed-model strategy (**ma\_5.2+5.1-codex-mini** with different deep/quick models) performed no better than same-model variants. Its +\$26 P&L is essentially flat. The multi-agent architecture bottleneck is structural, not model-dependent.

### 6. Ticker Type Matters More Than Strategy Choice

- **Predictable tickers** (ET at 63%, ZTS at 56%, SUPN at 54%): Stable businesses where HOLD is often correct. Most strategies perform similarly.
- **Unpredictable tickers** (RAPT at 10%, SNOW at 27%): High-volatility names where models consistently fail. RAPT (biotech micro-cap) is essentially random — 10% accuracy is worse than chance.
- **The winning strategy's edge comes from mid-difficulty tickers** (AMBA, SUPN) where active trading can exploit directional signals that exist but aren't obvious.

### 7. Actionable Recommendations

1. **Use single-agent with gpt-5.1-codex-mini** as the production strategy — it's simpler, cheaper (1 LLM call vs ~8-12), and more profitable.
2. **If using multi-agent, reduce the debate/risk layers** — the current consensus mechanism needs fundamental redesign. Consider: removing the risk manager veto, or requiring explicit BUY/SELL signals rather than defaulting to HOLD.

3. **Filter out micro-cap biotech tickers** (like RAPT) from automated trading — the data available to LLMs cannot predict binary event outcomes.
  4. **Focus on mid-cap stocks with clear fundamental stories** (like SUPN) where the model's analytical capabilities provide genuine edge.
  5. **Don't trust confidence scores for position sizing** — the most confident model (gpt-5.2, 0.64) underperforms the least confident (gpt-5.1-codex-mini, 0.37).
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## Methodology Notes

- **Outcome validation:** Each BUY/SELL decision is validated against actual market movement over the following 30-day window. A BUY is correct if price rose; a SELL is correct if price fell. HOLD is correct if price moved less than 1%.
- **P&L calculation:** Based on \$10,000 position size per trade, with 0.1% spread + 0.05% slippage transaction costs applied to all non-HOLD trades.
- **Signal extraction:** Decisions extracted from LLM text using regex pattern matching on "FINAL TRANSACTION PROPOSAL" lines, with fallback LLM extraction for ambiguous outputs.
- **Hard overrides:** Quantitative guardrails (RSI, SMA crossovers) can override LLM decisions when technical signals are extreme, though this was rarely triggered in practice.